



NEXUS 2.0

DecentralizedClaim – Autonomous AI & Blockchain Insurance Platform

Problem Statement: The Insurance "Black Box" Crisis

- **Lack of Transparency** 🔍
 - Users don't know why claims are approved/rejected
 - No visibility into decision-making
- **Slow Processing** ⌚
 - Claims take weeks due to manual workflows
 - Multiple handoffs across systems
- **Fraud & Inaccuracy** ⚠️
 - Systems cannot verify real-world data
 - Fraudulent claims increase costs for everyone



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DecentralizedClaim

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The World's First Autonomous Multi-Agent Insurance Resolution Engine

The Concept :

DecentralizedClaim is an end-to-end autonomous system that replaces slow human processes using a Directed Acyclic Graph (DAG) of 4 specialized AI agents.

- Uses agentic workflows + blockchain technology
- Converts opaque systems into transparent “glass-box” experience
- Designed for real-time, user-friendly claim resolution

Key Capabilities include:

✓ Processes claims in under **1 minute**

✓ Verifies claims using **real-world data** (weather, prices)

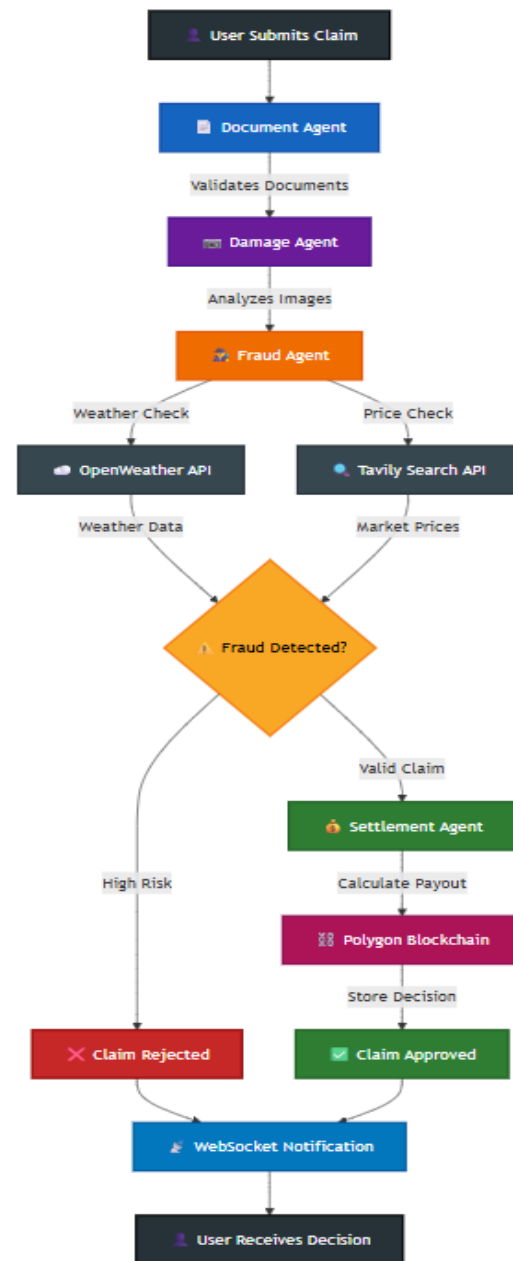
✓ Provides **transparent AI reasoning** in real-time

✓ Ensures **immutable decisions on blockchain**



TECHNICAL APPROACH

-  **Autonomous Workflow Automation:** Orchestrated a **Directed Acyclic Graph (DAG)** of 4 specialized AI agents using LangGraph (**Document, Vision, Fraud, and Settlement**) that independently debate and verify claim validity.
-  **Real-World Intelligence (MCP):** Through the **Model Context Protocol (MCP)**, they query live Weather APIs and Market-Price scrapers to verify claims against physical reality in real-time.
-  **Zero-Trust Transparency:** Every internal thought process of the AI is streamed live via WebSockets, and every final decision is cryptographically anchored to the **Polygon Blockchain** for permanent, public verification.
-  **Accessibility First:** By integrating **Voice-Native AI (Vapi)**, non-technical users can file complex claims through a natural conversation, eliminating the friction of traditional, daunting legal paperwork.





FEASIBILITY & VIABILITY



1. TECHNICAL FEASIBILITY

The system is **fully operational** today. It leverages **production-ready frameworks** (Next.js, NestJS, FastAPI) and isolates heavy AI workloads to **specialized microservices**, ensuring **high performance** and **scalability**.



2. COMMERCIAL VIABILITY

API execution costs (OpenAI + Tavily + Weather) per claim are **measured in cents**, compared to the **hundreds of dollars** it costs an insurance company to employ human adjusters for manual review.



3. DATA SECURITY & PRIVACY

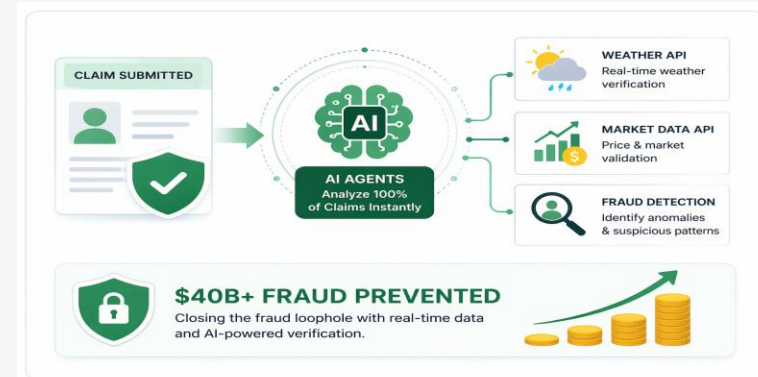
Utilizes a hybrid storage model. Sensitive user data and documents are kept in **encrypted Supabase relational databases** (GDPR compliant), while only **anonymized cryptographic hashes** are pushed to the public blockchain for verification.



IMPACT AND BENEFITS

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- **For Everyday People:** Transforms a highly stressful 3-week waiting period into a transparent, 2-minute settlement when they need funds the most. No more opaque "pending" statuses.
- **For the Industry:** Closes the \$40B+ fraud loophole. Instead of humans spot-checking claims, AI agents instantly cross-reference 100% of claims against real-world weather and market data APIs.
- **Restoring Trust:** The "Glass Box" UI and public blockchain verification portal allow anyone to independently audit claim outcomes, rebuilding trust between consumers and financial institutions.





REFERENCES AND TECH STACK

- **Chalise, A. (2024)** – Streamlining Insurance Claims using Agentic AI (LangGraph)
- **Blockchain Council (2025)** – Blockchain in Insurance Claim Processing (Zero-Trust Immutability)
- **Druid AI (2026)** – AI-driven Automated Claims Processing & Voice-Based Workflows
- **AutomationEdge (2026)** – RPA, Agentic AI & Zero-Touch Claims Automation (ROI & Scalability)

